



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context

In this capstone, I will take of the role of Data scientist for the new space company.

- Problems you want to find answers

We try to find insights from data of most successful space company (SpaceX) and prediction for our new space company.

Section 1

Methodology

Methodology

- Data collection methodology:

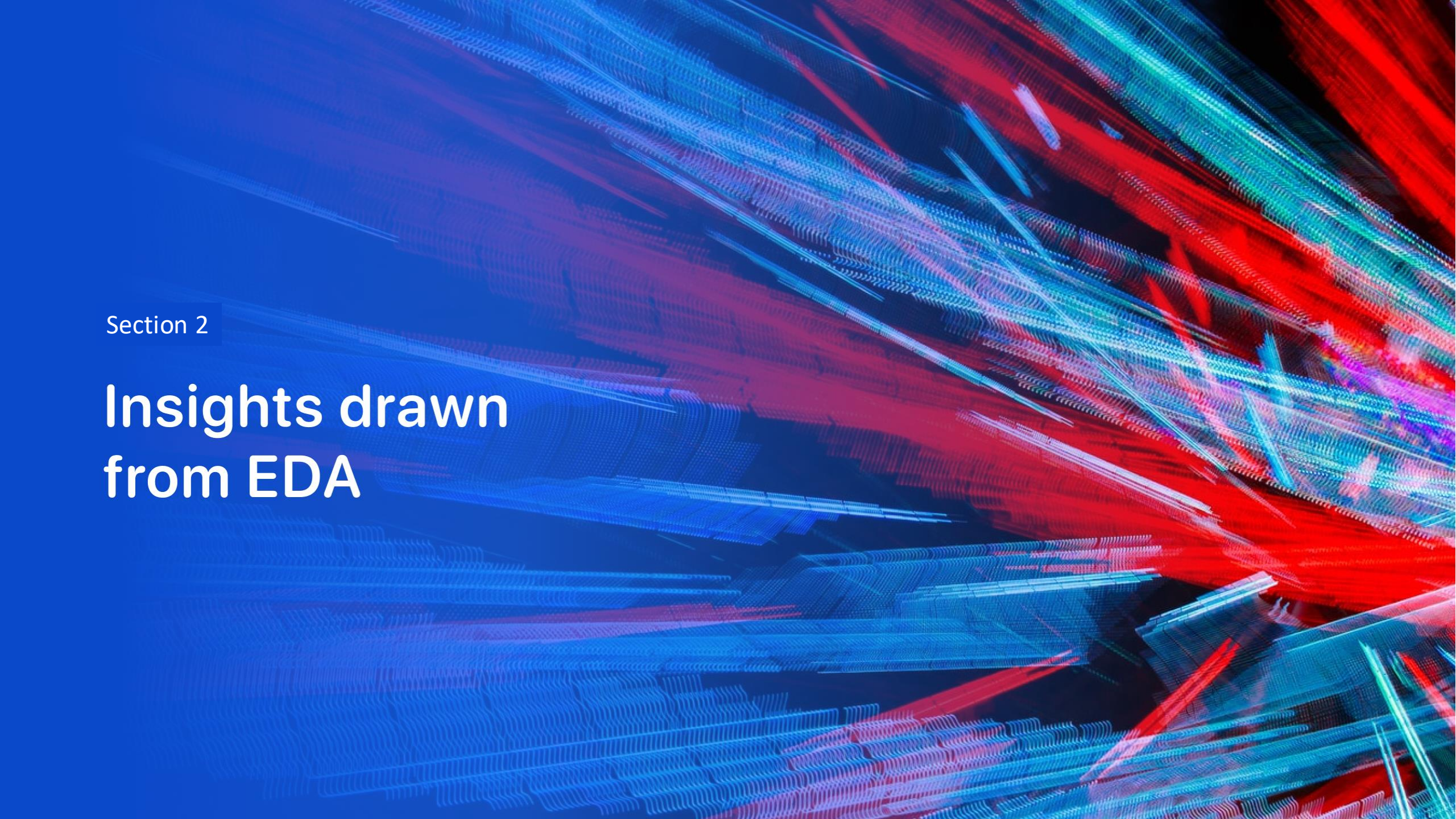
Collect data through SpaceX APIs and web scraping.

- Perform data wrangling:

We Create a landing outcome label from outcome column.

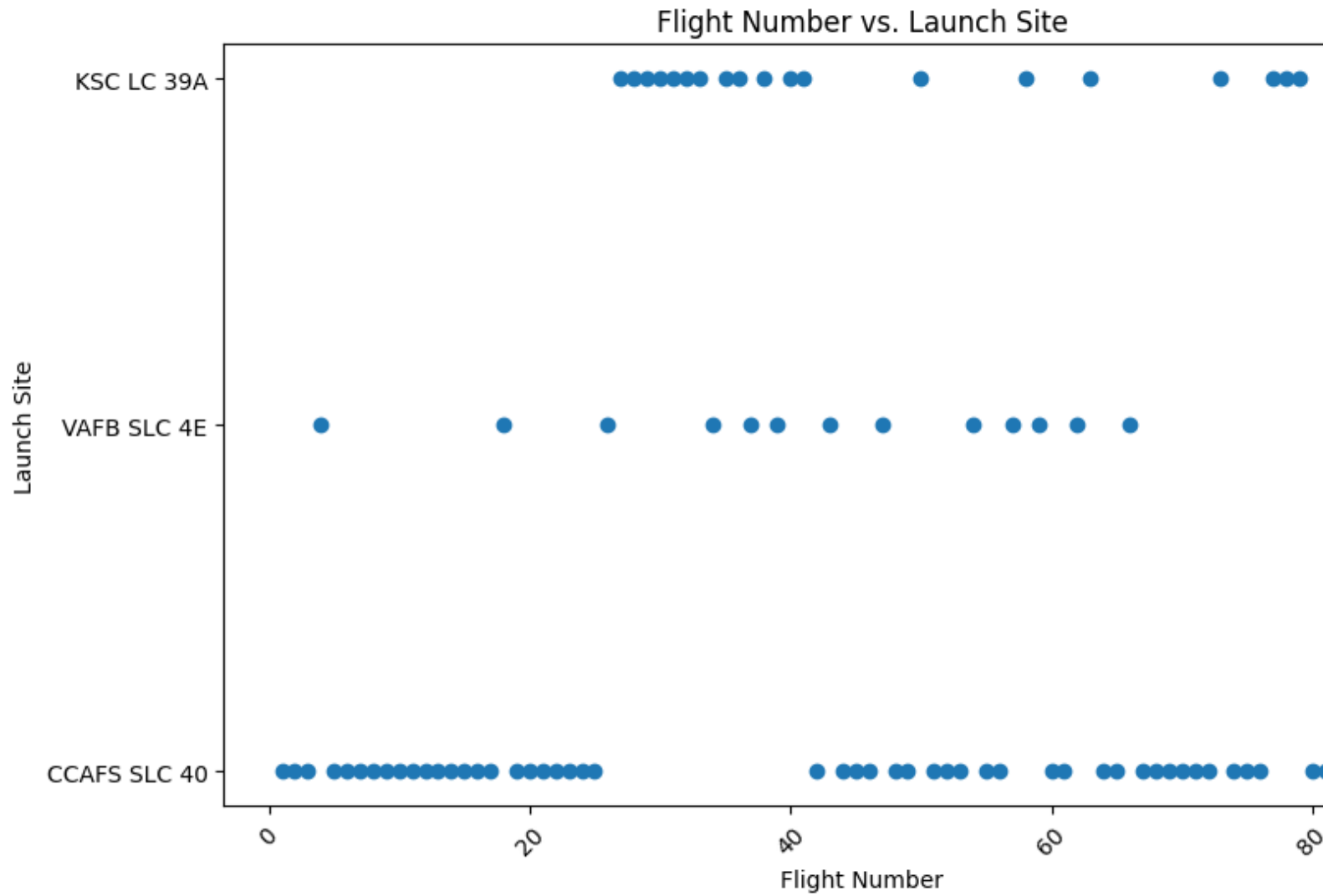
- Perform predictive analysis using classification model:

Here, we Train three model and make predictions from them and chose the best one.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

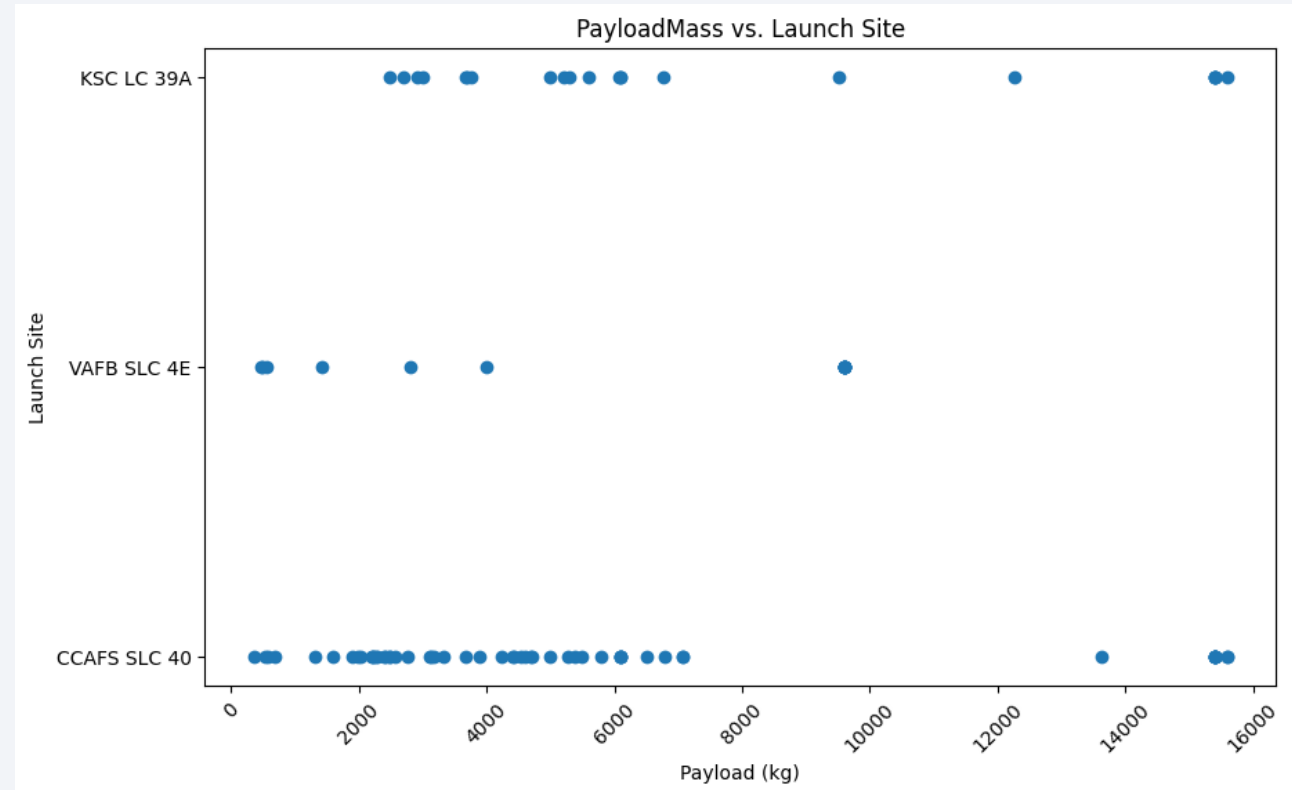


Flight Number vs. Launch Site

CCAFS SLC 40 launch site most of time used.

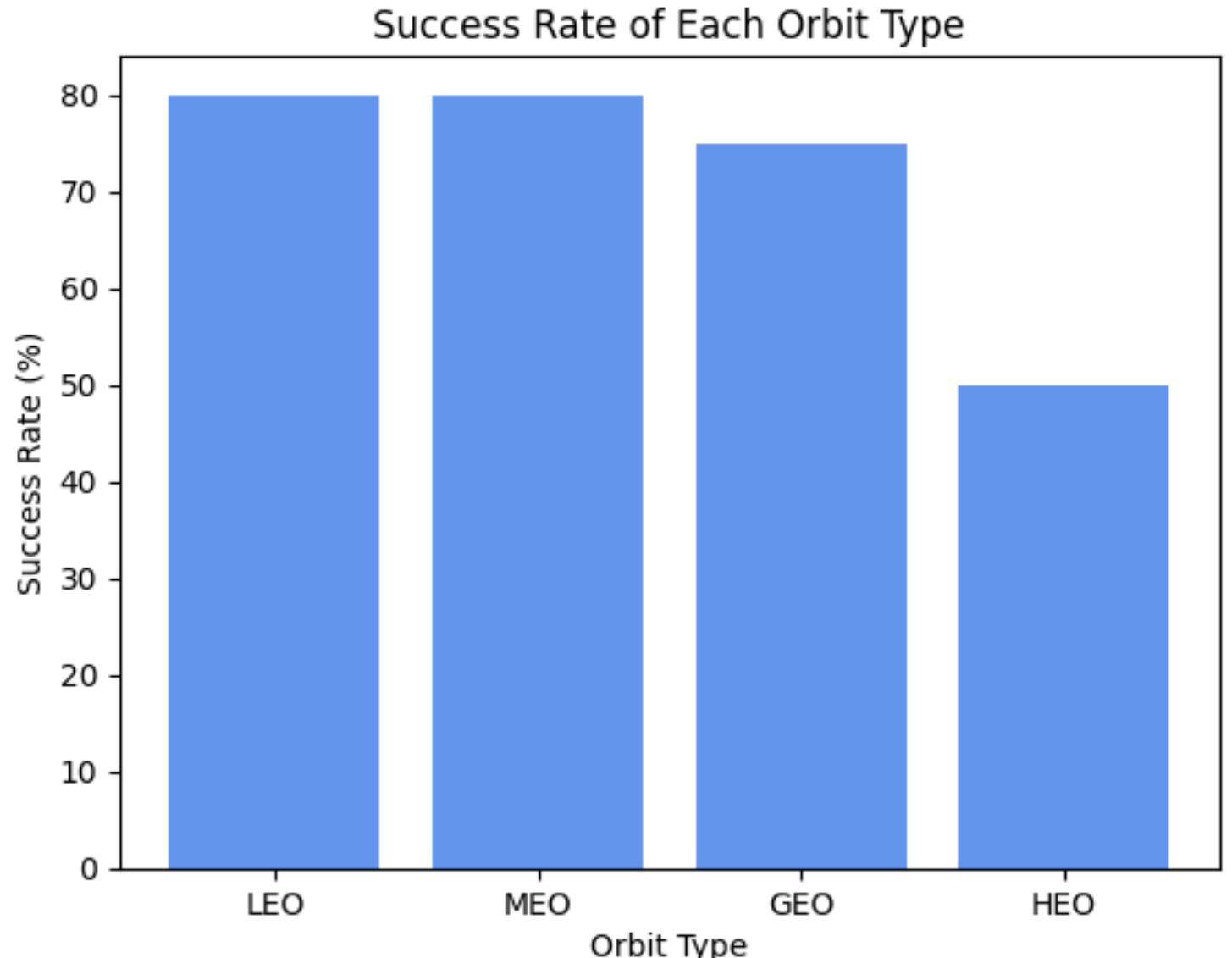
Payload vs. Launch Site

- Flight with Payloadmass more than 8000, Mostly taken By KSC LC 39A launch site.

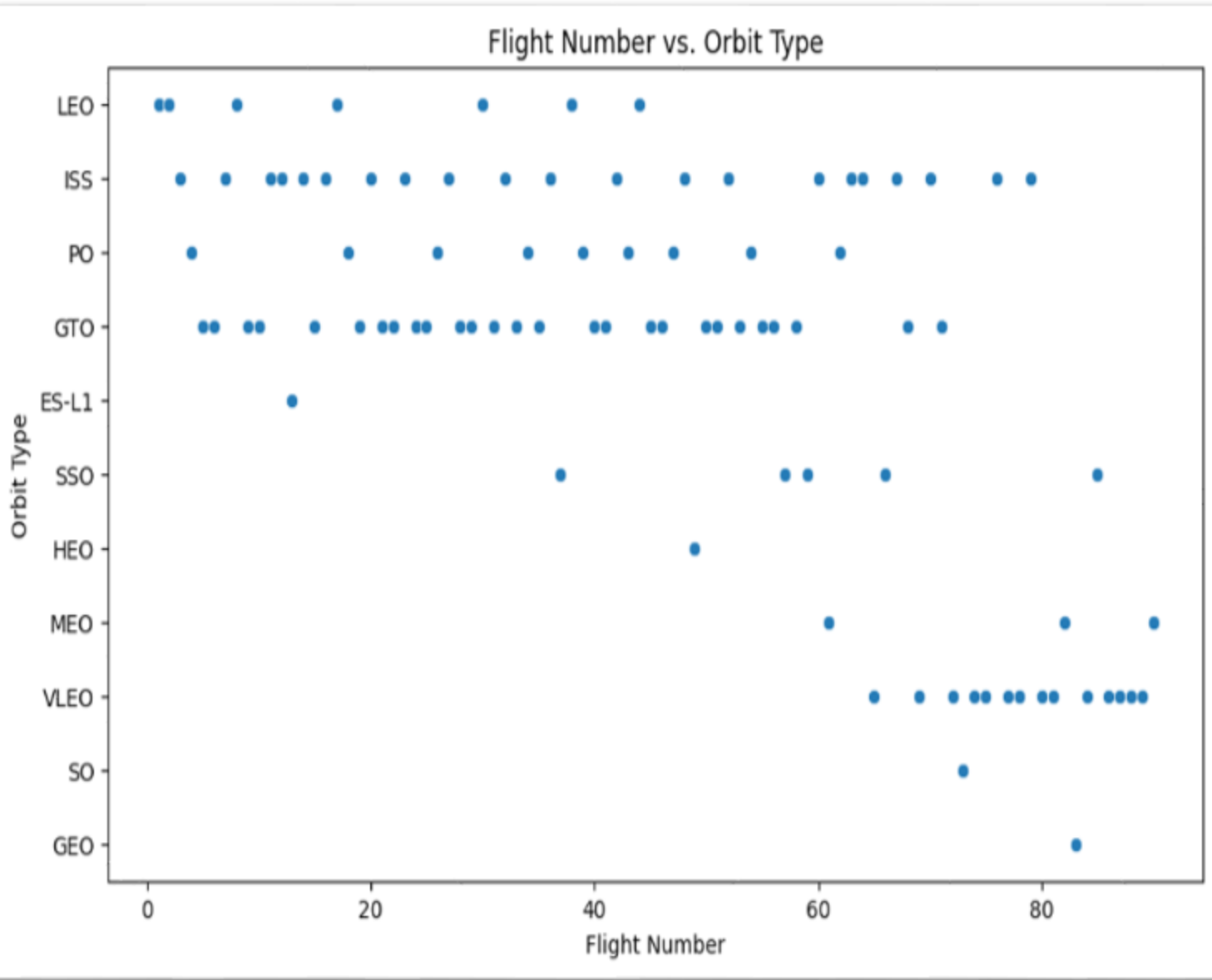


Success Rate vs. Orbit Type

This graph show that HEO orbit have minimum success rate.

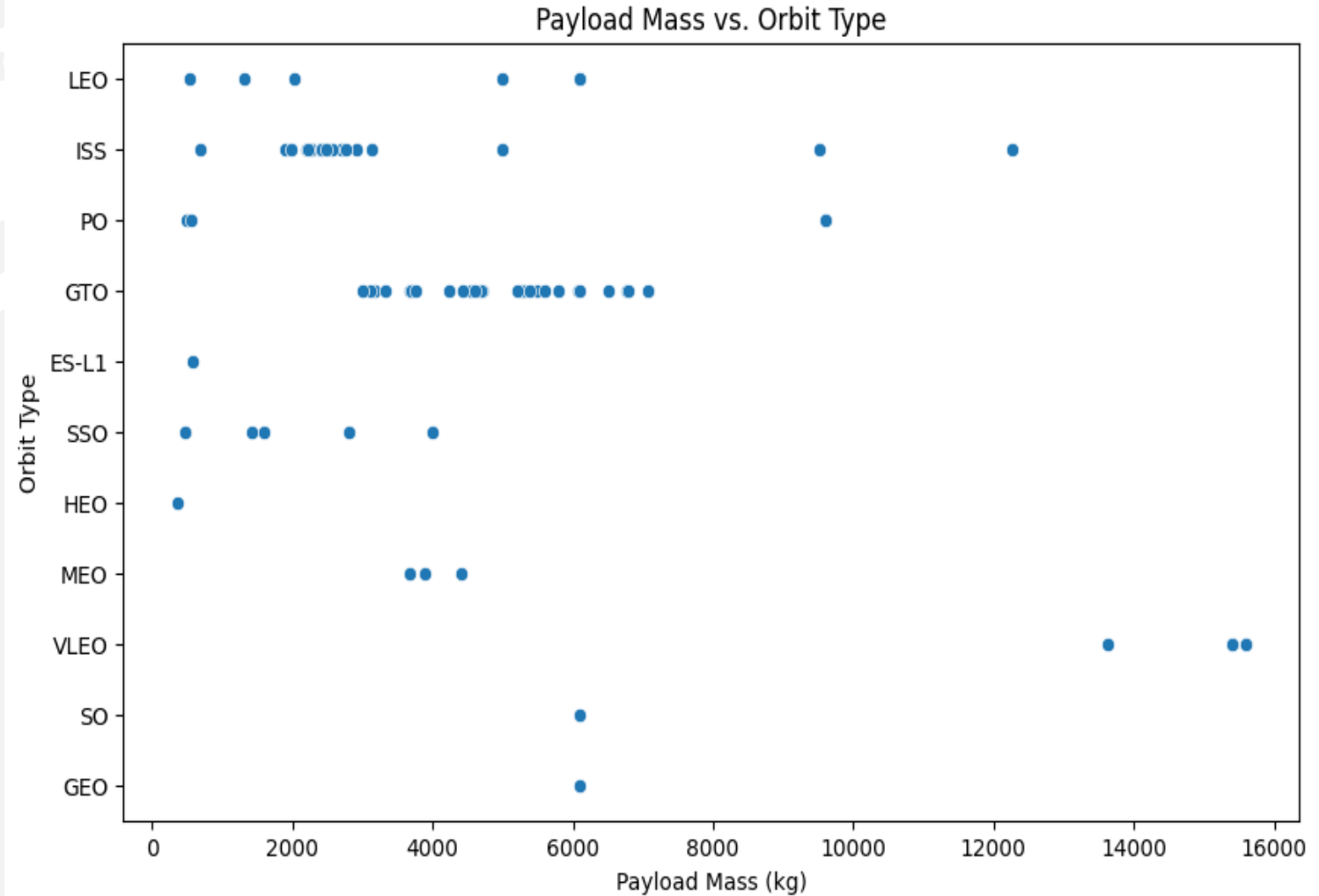


Flight Number vs. Orbit Type

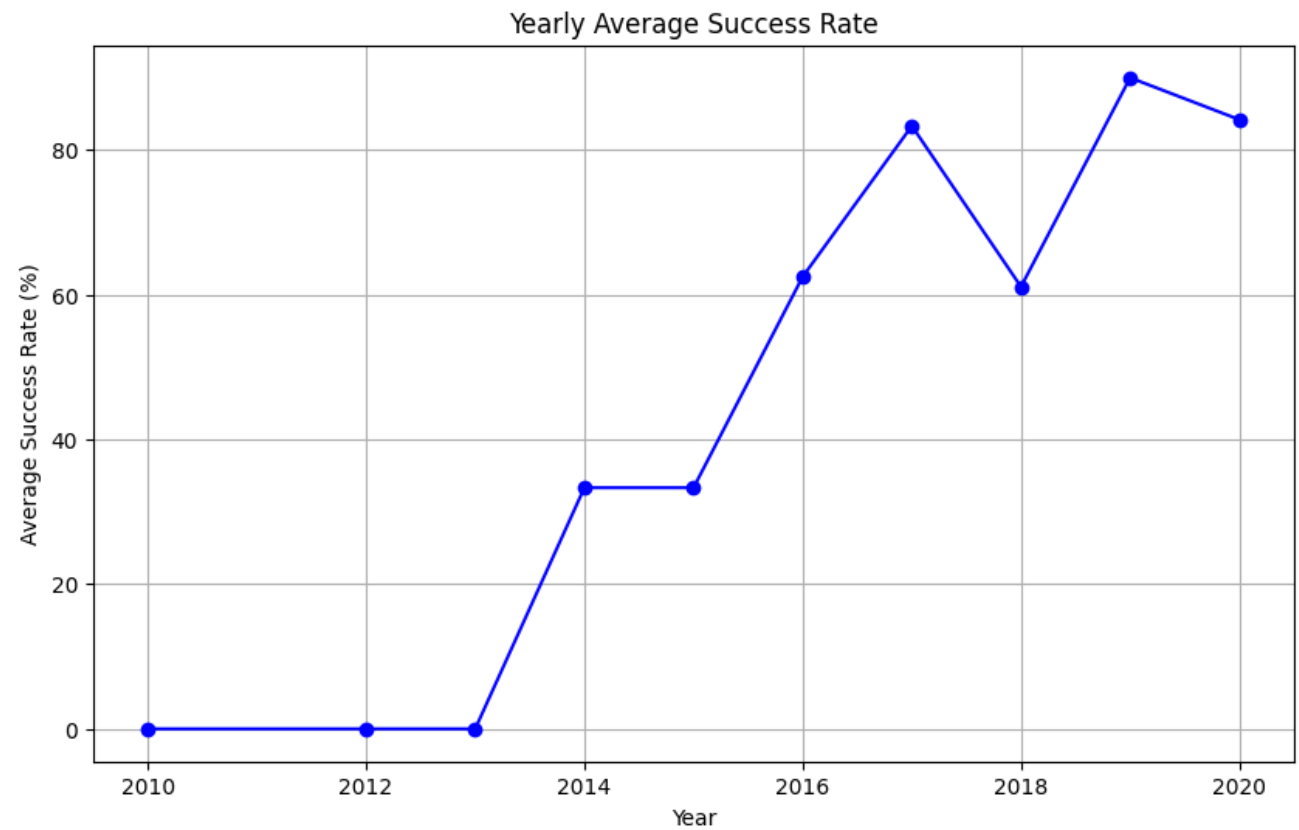


Payload vs. Orbit Type

VLEO orbit have maximum payload Mass.



Launch Success Yearly Trend



All Launch Site Names

'CCAFS SLC 40'

'VAFB SLC 4E'

'KSC LC 39A'

First Successful Ground Landing Date

- 2015-12-22

This is the first successful ground landing date.

Query(SQL):

```
SELECT Date  
FROM launch_records  
WHERE Outcome = 'True RTLS'  
ORDER BY Date ASC  
LIMIT 1;
```


Successful Drone Ship Landing with Payload between 4000 and 6000

- Only **Falcon 9** booster that have successfully landed on a drone ship (**True ASDS**) with a payload mass between 4000 and 6000 kg.

- Query(SQL):

```
SELECT BoosterVersion  
FROM launch_records  
WHERE Outcome = 'True ASDS'  
AND PayloadMass > 4000  
AND PayloadMass < 6000;
```

Total Number of Successful and Failure Mission Outcomes

- After filtering the data for outcomes that indicate success or failure, we find there are 60 successful missions and 9 failed missions in the dataset.

Query(SQL):

-- For Successful Missions

```
SELECT COUNT(*) AS successful_missions  
FROM launch_records  
WHERE Outcome LIKE '%True%';
```

-- For Failed Missions

```
SELECT COUNT(*) AS failed_missions  
FROM launch_records  
WHERE Outcome LIKE '%False%';
```

Boosters Carried Maximum Payload

- Explanation:
- The **BoosterVersion** is consistently **Falcon 9**, which carried the payload mass of **15,600 kg**.
- This payload mass appears three times in your dataset (Flight numbers 68, 73, and 76).
- Therefore, **Falcon 9** is the booster that has carried the maximum payload of **15,600 kg** across multiple flights.

Query(SQL):

```
max_payload_row = df[df['PayloadMass'] == df['PayloadMass'].max()]
```

```
# Display the BoosterVersion associated with the maximum payload
```

```
print(max_payload_row[['BoosterVersion', 'PayloadMass']])
```

2015 Launch Records

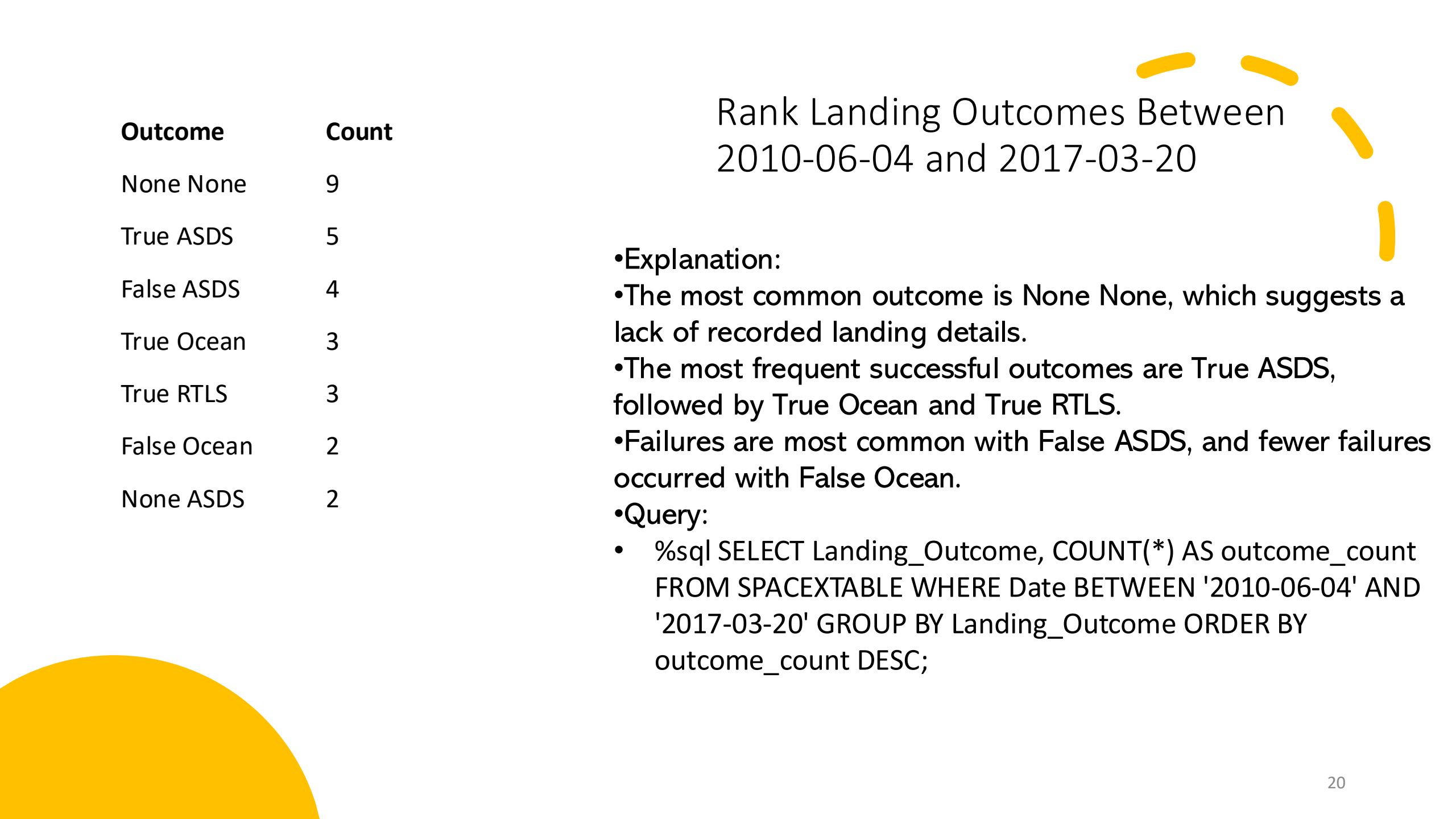
- There were **two failed drone ship landings** (False ASDS) in 2015 involving **Falcon 9** from the **CCAFS SLC 40** launch site.

- Query(SQL):

```
max_payload_row = df[df['PayloadMass'] == df['PayloadMass'].max()]
```

```
# Display the BoosterVersion associated with the maximum payload
```

```
print(max_payload_row[['BoosterVersion', 'PayloadMass']])
```



Outcome	Count
None None	9
True ASDS	5
False ASDS	4
True Ocean	3
True RTLS	3
False Ocean	2
None ASDS	2

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Explanation:

- The most common outcome is None None, which suggests a lack of recorded landing details.

- The most frequent successful outcomes are True ASDS, followed by True Ocean and True RTLS.

- Failures are most common with False ASDS, and fewer failures occurred with False Ocean.

- Query:

- ```
%sql SELECT Landing_Outcome, COUNT(*) AS outcome_count
FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND
'2017-03-20' GROUP BY Landing_Outcome ORDER BY
outcome_count DESC;
```



A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

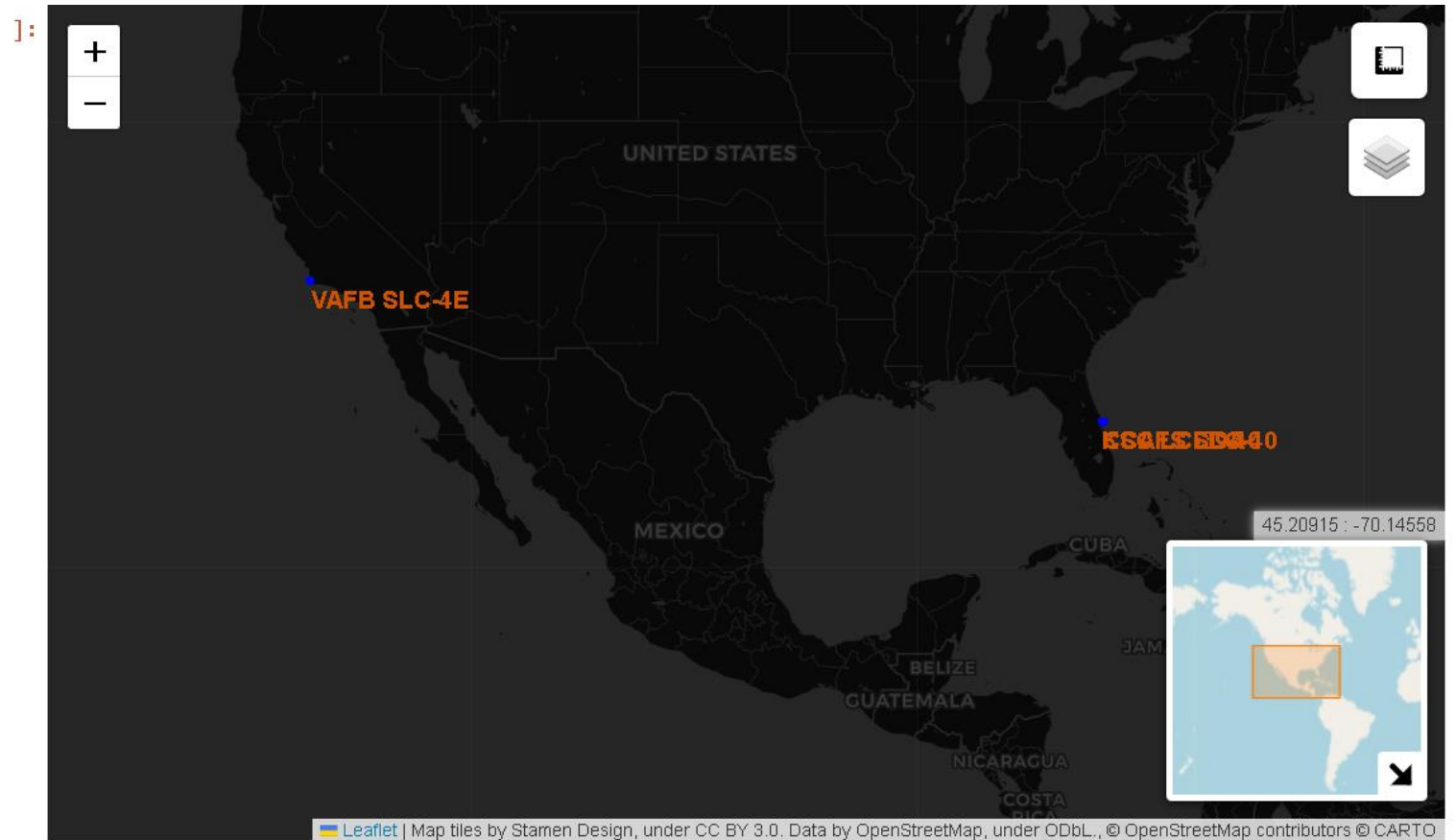
Section 3

# Launch Sites Proximities Analysis

# Location of Launch Sites

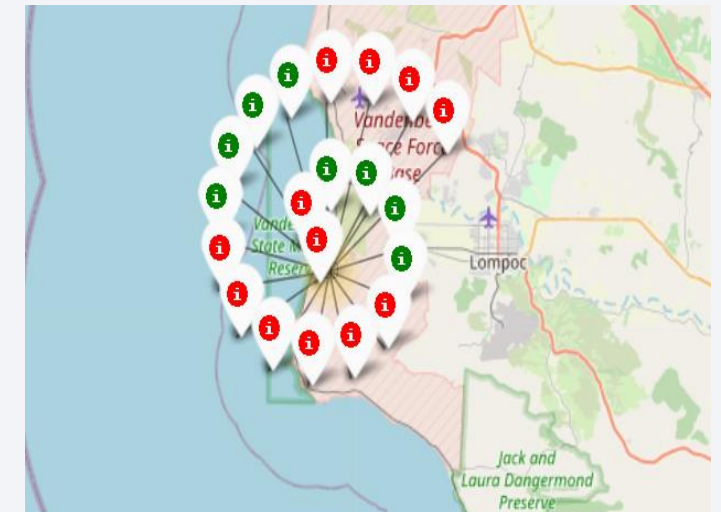
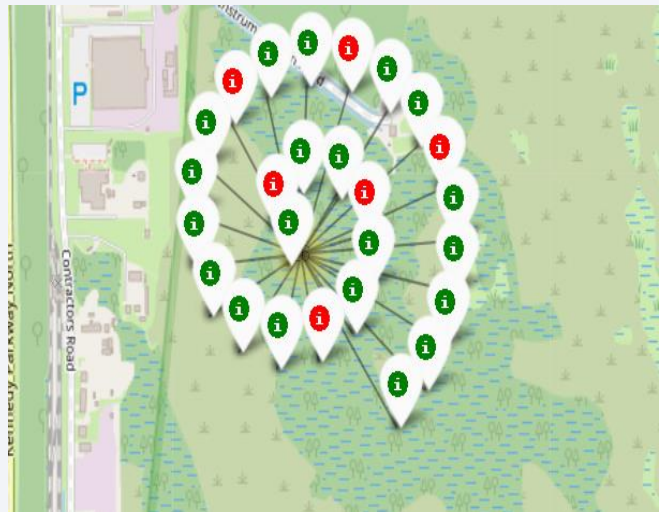
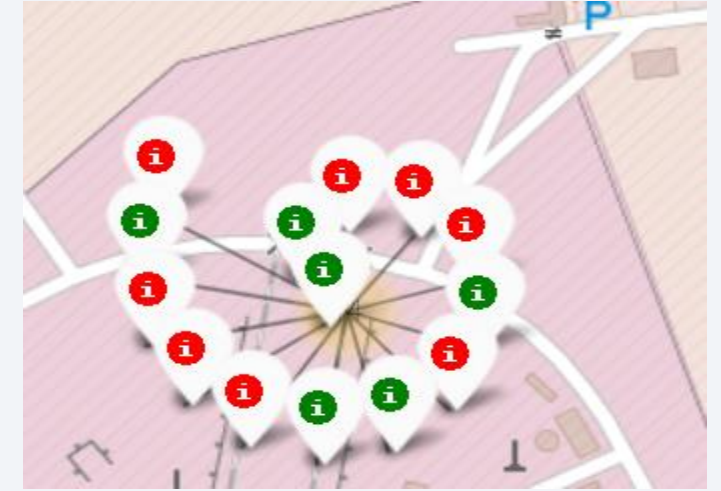
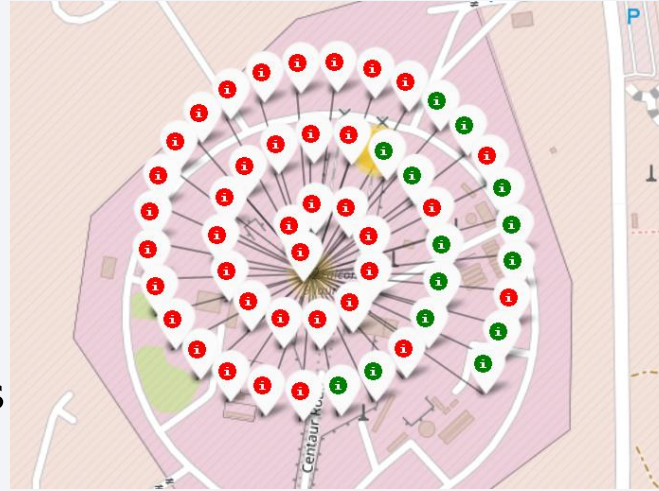
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We see clearly all locations are near the cost and proximity to the equator line.



## color-labeled launch outcomes

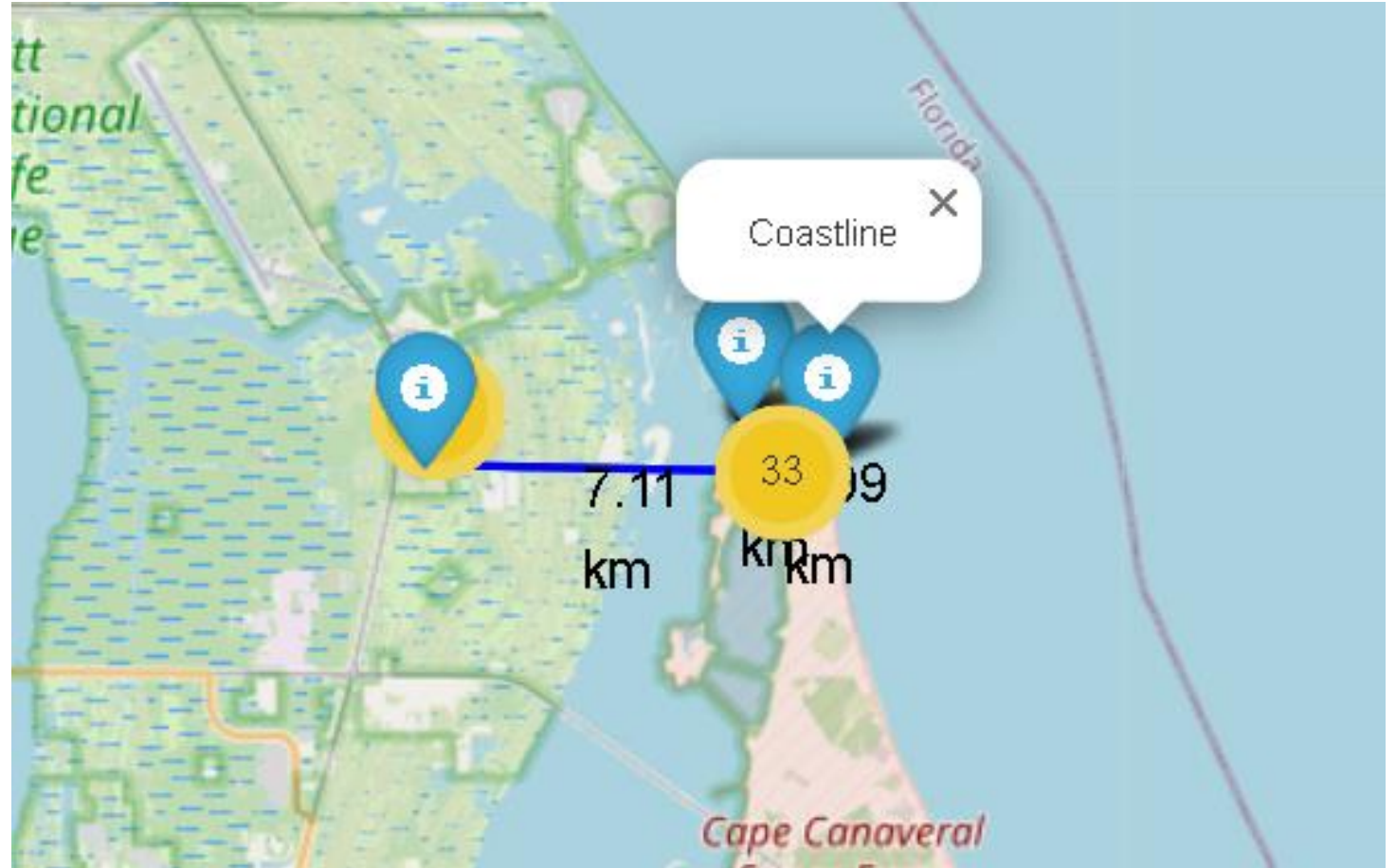
The map will display clustered markers at the launch site locations. As you see in, the clusters will expand, and you can see individual markers with colors indicating success (green) or failure (red).





## Calculate distances:

- **Markers** for the launch site and nearby proximity features like railway, highway, and coastline.
- **Lines** connecting the launch site to each proximity.
- **Distance markers** in kilometers showing the distance between the launch site and each proximity.







Section 5

# Predictive Analysis (Classification)

# Classification Accuracy

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- Visualize the built model accuracy for all built classification models, in a bar chart
- Find which model has the highest classification accuracy



# Confusion Matrix

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- Show the confusion matrix of the best performing model with an explanation

# Conclusions

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- Point 1
- Point 2
- Point 3
- Point 4
- ...

# Appendix

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- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

